

S.4 HOLIDAY WORK

Respond to **all** the items in this section and paper.

Item 1

Kyle's house is enclosed by a perimeter wall. His bathroom mirror labelled **mirror A**, makes one appear bigger while using it, especially when a person stands close to it. He also has some pieces of mirror labelled **mirror B**, in which the size of the image is the same as that of the object, no matter how far an object is placed in front of it.

The planning authority in Kyle's area passed new guidelines requiring that all perimeter walls less than 10 m away from the roadside must be broken down. Kyle's perimeter wall near the road seems to be in danger of being broken down. Kyle lacked any measuring instrument to establish the actual distance of that wall from the roadside. Kyle's neighbour across the road advised him to use echo method. His neighbour's wall is 60 m from the side of the road opposite that of Kyles residence. The road is 8 m wide. Kyle stood alongside his perimeter wall that is close to the road and clapped two pieces of wood, once. He heard the reflected sound from the neighbour's wall 9.5 seconds later. However, Kyle and his neighbour did not know how to use the information obtained.

Of late, some young men have been consistently converging along the road near Kyle's perimeter wall. He wants to be watching them closely without the young men seeing him, but Kyle lacks installed security cameras at his residence.

Kyle has contacted you for help.

Hint:

- The speed of sound in air is 330 ms^{-1} .
- Kyle's perimeter wall near the road, the road and the perimeter wall of the neighbour across the road are parallel to each other.

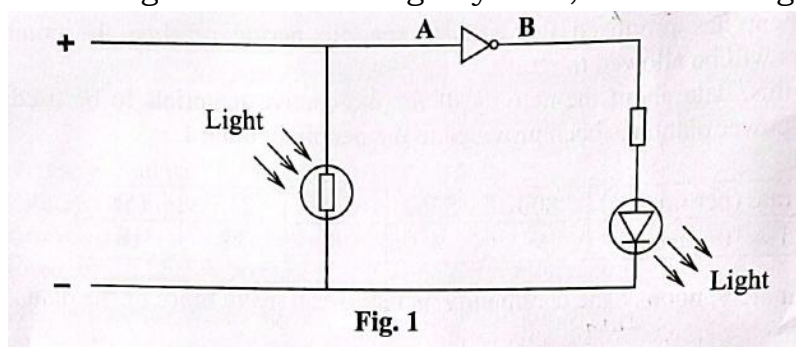
Task:

- a). Help Kyle to:
 - i. understand why images in **mirror A** appear as observed (include an illustration).
 - ii. determine whether his perimeter wall will be among those to be broken down.
- b). Describe how Kyle can design an optical instrument using two pieces of **mirror B** to watch the young men closely (include an illustration).

Item 3

During dry season, a sailor in an ocean embarked on a journey to transport fish to a certain island for packaging. The sailor was not sure of the direction to take in order to reach the destination (Island) safely without getting lost. The sailor contacted her friend who was already at the destination. In a bid to help the sailor, her friend sent a *Google map* link and guided the sailor on how to use it. The sailor followed it up to reach the destination (Island) without getting lost. However, the sailor did not understand how the google map works.

On arrival at the destination (Island), the sailor noticed that the level of water started rising in the ocean at night; where the waves carried large volumes. This could cause any sailor's boat to become unstable and eventually capsize. The sailor wondered whether there are any benefits of such rising water levels, to the communities living around such water bodies, during dry season. While at the Island, the sailor also noticed that, the Island is equipped with an automatic security. light system that switches on at night and off during day time, as seen in figure 1.



The sailor was also amazed by the automatic lighting system.

The sailor wanted to be educated on these occurrences or observations made.

You have been contacted to be of help to the sailor.

Task:

- Help the sailor to understand:
 - why the water behaved the way it did.
 - how the application sent by her friend works.
- Educate the sailor on how the security light system in the circuit, in figure 1 works (include truth table).
- Explain any **two** benefits the sailor needs to know about the water behaviour observed.

SECTION D

Part I

Respond to **ALL** items from this part.

Item 4

Abdu always rides a bicycle to school 10 km away from his home. One morning he packed tea in a metallic bottle and departed for school at 7:00 am. Abdu noticed that when he pedals the bicycle, whether travelling at low speed or not, it makes noise which is persistent and discomforting. He did not know why his bicycle behaved this way.

After covering 6 km, Abdu noticed that it was already 7:40 a.m., meaning that he would reach late. This brought about the need to accelerate so that he could arrive at school on time. Abdu increased his velocity from 5 ms^{-1} to 15 ms^{-1} and was able to cover the remaining distance to school to arrive on time. However, when Abdu's classmates asked him for his rate of change of velocity he completed the journey with, he had no idea.

At lunch time, Abdu took the tea he had packed and discovered that the tea was only warm, and was not that enjoyable at all. His friend advised him to buy a vacuum flask to keep the tea hot for a longer period of time. However, Abdu did not know how the flask could keep the tea that hot as stated.

Abdu also wondered how the processes used to minimise heat loss in the device could be used in other aspects in real life.

Abdu has reached out to you for guidance.

Task:

- a) Help Abdu understand the likely cause of the discomfort he experienced while traveling to school and how it could be minimised.
- b) Determine Abdu's rate of change of velocity, so that he can respond to his classmates.
- c) With the aid of a diagram, help Abdu to understand how his friend's advice will solve the problem he faced at lunch time.
- d) Explain any **one** practical application of each of the processes the device uses to keep the liquid as desirable.

Item 5

Students from a certain school had a trip to the hydroelectric power plant. The person in charge at the plant took them around. They at first went to see the wall of the dam. The person in charge told the students that the wall of such a dam is always built thicker and wider at the bottom than at the top. The students did not know why the wall of the dam should be built that way and feared to ask the in charge, yet they were eager to know.

Later, the person in charge took the students to a nearby construction site. At the site, a mixture of sand, gravel, cement, water and steel bars was being made to erect columns for a storeyed building. The person in charge told the students that each of the steel bars being used are of length 3.2 m, diameter of 2.5 cm, and would stretch by 0.2 mm when a tensile force of $7.7 \times 10^5 \text{ N}$ is applied. The students wondered whether the steel bars met the requirements for use in the structure, given the heavy weight of the building.

After lunch, the students visited a beach and noticed that while standing out of water, they felt a cool breeze compared to when they were sailing in a boat and wondered why the difference. One of the students quickly mentioned that there could be a number of applications of the breeze they experienced, in real-life .

Majority of the students did not know any practical applications of the breeze they experienced, since they had not yet learnt but were interested in finding out.

On returning from the trip, the students approached you for help.

Hint ,

Steel bars are of good quality if the ratio of stress to strain exceeds $2.1 \times 10^{11} \text{ Nm}^{-2}$.

Task:

- a) Explain to the students why the wall they observed was designed as they observed.
- b) Help the students to determine whether the steel bars were of good quality.
- c) Explain to the students any two practical applications of the breeze they experienced.

Part II

Respond to **one** item from this part.

Item 6

A business person owns a company that uses current electricity to operate machines and other appliances. One day, as the workers in the company were switching on the lights, they noticed that all the bulbs were dim. Subsequently, the business person called an electrician to assess the problem, who later established that the mains supply of 415 V was low and needed to be increased to 2,000 V.

The electrician assured the business person that for the problem to be solved, the business person must buy an efficient device such that the power loss does not exceed 5 W. The business person does not have any information on practical ways of how to minimise energy losses in the device.

You have been contacted by the business person for help.

Hint

Cables of resistance 2Ω supply 2 kW of electrical power, if it is transmitted at 2,000 V.

Task

- a) Help the business man understand the design and mode of operation of the device needed to sort out the voltage issue .

- b) Determine whether the device is efficient, so that you can educate the business man.
- c) Describe to the business person any **two** practical ways of minimising energy losses in the device.

Item 7

A business person plans to set up a new restaurant which has only one socket for the mains. The business person wants to connect the items shown in table 2, for use.

Table 2

S/N	Item	Power Rating
1	Loudspeaker	240 V, 60 W
2	Electric Kettle	240 V, 1 kW
3	Torch	12 V, d.c.
4	T.V	12 V, d.c.

However, the business person does not know how they can be connected. In order to budget properly, the business person wants to know the cost of electrical energy needed for operating a loudspeaker for 10 hours a day per week. The business person was told that it is possible to pay less than the expected amount if practical steps are taken to save energy. However, the idea of saving energy sounded new to the business person and wanted to be educated about it.

The business person was also told that there are some missing electrical components needed to be connected with the items in table 2. It was not easy to tell the missing components.

You have been contacted by the business person to offer guidance.

Hint

The unit cost of electrical energy is Shs600.

Task:

- a) Help the business person to:
 - i. identify the missing components required for the items in table 2 to be connected safely.
 - ii. explain the purpose of each component.
 - iii. illustrate, using a circuit diagram, how the devices in table 2 can be connected with the identified components.
- b) Find out how much money is needed to operate the loudspeaker as planned, so that the business person can budget accordingly.
- c) Explain any **two** practical steps the business person can take, in order to minimize energy losses to save energy and pay less for energy consumed .

END